QPCODE

Reg. No:.....

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025

THIRD SEMESTER

DSE - MG3DSEPHY200- Basic Semiconductor Physics

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours

Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any 10 questions

Each question carries 2 marks

1. Discuss how a barrier potential is developed at the PN Junction ? [U] / [CO1]
2. Define junction capacitance and diffusion capacitance. [K] / [CO1]
3. Differentiate between ideal diode and real diode. [U] / [CO2]
4. What is a zener diode ? Draw the schematic symbol for a Zener diode . [K] / [CO2]
5. Write an expression for diode current and explain the symbols used in it. [K] / [CO2]
6. Discuss the terms efficiency and ripple factor of a rectifier. [U] / [CO3]
7. What is a filter circuit? What is its function? [K] / [CO3]
8. Give the circuit diagram of a half wave rectifier and illustrate its input and output waveforms. [U] / [CO3]
9. What is meant by Q point of a transistor? [U] / [CO4]

10. For a transistor $\alpha = 0.95$ and $I_E = 1mA$, find the values of I_C and I_B. [A] / [CO4]
11. What are the advantages of negative feedback? [K] / [CO5]
12. Draw the block diagram of voltage-series and voltage-shunt feedback. [U] / [CO5]
13. Draw a voltage divider bias circuit for CE Amplifier. [K] / [CO5]
14. What is an oscillator? [U] / [CO6]

[2 x 10 = 20]

Part B

Short Essay Questions

Answer any 6 questions

Each question carries 5 marks

15. Explain how the process of avalanche breakdown occurs ? [U] / [CO1]
16. Define zener voltage ? How does a zener diode maintain a constant voltage across it ? [U] / [CO2]
17. Explain the working of a shunt capacitor filter and give the expression for ripple factor of the same. [U] / [CO3]
18. In a transistor circuit, collector load is 4KΩ whereas quiescent current is 1mA. (i) what is the operating point if Vcc = 10V? (ii) what will be the operating point if Rc = 5KΩ? [A] / [CO4]
19. With the help of a connection diagram draw and explain the output characteristic of a common- base configuration of a transistor. [U] / [CO4]
20. The voltage gain of an amplifier without feedback is 3000. Calculate the voltage gain of the amplifier if negative feedback is introduced in the circuit. [A] / [CO5]
21. Derive the expressions for Current, Voltage and Power gains. [U] / [CO5]
22. Explain the principle and working of Hartley Oscillator with the help of a circuit diagram. [U] / [CO6]

[5 x 6 = 30]